

Question Paper

Exam Date & Time: 30-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.Tech. (INFORMATION TECHNOLOGY) EXAMINATIONS - DECEMBER 2024 / JANUARY 2025
SUBJECT: ICT 3123/ICT_3123 - SOFTWARE ENGINEERING
(MAKEUP)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Evaluate the strengths and limitations of the Concurrent Specialized Process Model when applied to large-scale, distributed software development projects. How does this model work in comparison to the Waterfall and Agile models in terms of flexibility, scalability, and risk management? Provide specific examples to support your assessment. (5)
- 1B) TechSol, a software development company, is developing a cloud-based customer management system (CMS) for a client. The project faces tight deadlines, and the team is working with an unfamiliar technology stack. Additionally, the client requires the product to be deployed across multiple regions, each with its own compliance requirements. The project manager has highlighted several risks that may impact the success of the project. Based on this scenario, identify and classify at least six potential risks according to the nature of their impact on the project. Explain how the identified risks could jeopardize the success of the CMS project. (3)
- 1C) Consider an online shopping system where customers can search for products, add them to a cart, and proceed to checkout. The thread-based testing identifies the following classes involved in the checkout process: (2)
- Cart Class: Manages the items that the user has added to their shopping cart.
 - Payment Gateway Class: Handles the payment processing for the order.
 - Order Confirmation Class: Generates the order confirmation and updates the order status.
- Describe how you would integrate and test these classes to ensure that the checkout functionality works correctly when a user places an order. Provide at least two specific test cases for this thread.
- 2A) Create a class diagram for the management of flights and pilots scenario given below: (5)
- An airline operates flights. Each airline has an ID. Each flight has an ID a departure airport and an arrival airport: an airport as a unique identifier. Each flight has a pilot and a co-pilot, and it uses an aircraft of a certain type; a flight has also a departure time and an arrival time. An airline owns a set of aircrafts of different types. An aircraft can be in a working state, or it can be under repair. In a particular moment an aircraft can be landed or airborne. A company has a set of pilots: each pilot has an experience level: 1 is minimum, 3 is maximum. A type of aeroplane may need a particular number of pilots, with a different role (e.g.: captain, co-pilot, navigator): there must be at least one captain and one co-pilot, and a captain must have a level 3.
- 2B) Discuss the differences between elicitation and specification in requirements engineering. Highlight the key challenges encountered during elicitation and explain how these are addressed in the specification process. (3)
- 2C) Consider a software application consisting of five modules (A, B, C, D, and E) with the following dependencies: (2)
- Module A calls Module B and Module C.

- Module B calls Module D.
- Module C calls Module E.
- Module D has no further dependencies.
- Module E is a standalone module.

- a) Determine the number of stubs required if you want to test Module A in isolation.
 b) Identify the number of drivers needed to test Module D independently.
 c) Provide the reason for the number of stubs and drivers considered in the above cases (a & b) .

3A) You have been tasked with designing the software architecture for a large-scale e-commerce platform. The platform needs to handle various functionalities, including managing customer orders, processing payments, maintaining product catalogs, and providing real-time updates for inventory and delivery statuses. The system also needs to ensure high availability, support for real-time data processing, and scalability for millions of users. (5)

Given the requirements of the e-commerce platform, evaluate which architectural style mentioned below would be most suitable for this system: Data-centered architecture, Data-flow architecture, Object-oriented architecture, or Layered architecture.

Justify your choice and explain the chosen architecture style.

3B) A mobile banking application requires users to log in using a 6-digit PIN. The system checks if the PIN is valid based on specific rules: (3)

- The PIN must contain exactly 6 digits.
- It must only contain numeric characters.
- The PIN cannot be sequential (e.g., 123456 or 654321).
- The output will either be "Valid PIN" or "Invalid PIN" based on these rules.

Using Black Box Equivalence Class Testing, define and describe the input and output equivalence classes that should be tested for this scenario. For each class, provide an example test case, and explain why it is important to consider that class.

3C) A leading healthcare organization is developing a new electronic health record (EHR) system aimed at enhancing patient care and streamlining administrative processes. The system is expected to facilitate patient data management, allow secure access for healthcare professionals, support clinical decision-making, and comply with stringent regulatory standards. Key stakeholders include doctors, nurses, administrative staff, and patients. The organization emphasizes the importance of data security, user satisfaction, system interoperability, and the ability to analyze large datasets for insights into patient outcomes. (2)

In the context of the EHR system classify the various quality attributes that are essential.

4A) Draw the control flow graph for the following function named find maximum and determine its cyclomatic complexity: (5)

```
int find-maximum(int i,int j, int k){
  int max;
  if(i>j) then
    if(i>k) then
      max=i;
    else max=k;
  else if(j>k)
    max=j
  else max=k;
  return(max);
}
```

4B) Imagine you are part of an Agile team developing a new mobile app. After releasing an early prototype to users, you receive feedback that certain features are confusing and not user-friendly. How would you and your team use Agile principles to respond to this feedback in the next development sprint, and what steps would you take to ensure that the final product meets user expectations (3)

4C) Identify the purpose of architectural patterns in software design. (2)

5A) Draw UML sequence diagram for chat applications for exchanging messages in chat room. (5)

5B) Identify and explain the main activities involved in the **requirement engineering** process. How do these activities ensure that the final product meets user needs? (3)

- 5C) Identify four common risks associated with software development projects. For each risk, briefly explain a possible mitigation strategy that could be employed to minimize its impact. (2)

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